

# ChinookAnalytics: Data Discovery Report

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February 18, 2026

## Abstract

This document summarizes what data is available in *ChinookAnalytics*, how it is structured, and what kinds of business analyses it can reliably support. It is designed to be readable by business stakeholders while remaining technically auditable.

All metrics were computed from SQL profiling executed against the curated **core** schema. (Profiling SQL: `sql/07_data_profiling.sql`.)

## 1 Executive Summary

**Domain.** A simplified digital music store: customers purchase tracks; tracks belong to albums and artists; tracks have genre/media attributes.

### Coverage.

- Transactions from 2021-01-01 to 2025-12-22 (5 years; 60 months)
- 412 invoices, 2240 invoice lines, 2328.60 total revenue
- 59 customers across 24 countries

**Reliability.** Primary identifiers are unique; critical transaction fields are non-null; and audited relationships have zero missing references.

### Important limitations.

- All customers have 6–7 invoices (no one-time buyers) → weak for churn/retention
- Invoice line `quantity` is constant (=1) → revenue driven by product mix, not basket scaling
- No discount/promotion fields → cannot measure promo impact

## 2 What Data Exists (Business View)

A purchase is represented by:

- **Invoice:** purchase event (date, billing country, total).
- **InvoiceLine:** items inside an invoice (track, unit price, quantity).

Products are structured as:

Artist → Album → Track

Tracks also include **Genre** and **MediaType**.

This enables analysis by:

- **Who:** customer
- **When:** invoice date (year/month)
- **Where:** billing country
- **What:** artist/album/track/genre/media type

### 3 Core Schema Overview (Technical Summary)

#### 3.1 Row Counts

| Table              | Rows |
|--------------------|------|
| core.customer      | 59   |
| core.employee      | 8    |
| core.artist        | 275  |
| core.album         | 347  |
| core.track         | 3503 |
| core.genre         | 25   |
| core.mediatype     | 5    |
| core.invoice       | 412  |
| core.invoiceline   | 2240 |
| core.playlist      | 18   |
| core.playlisttrack | 8715 |

Table 1: Row counts in **core**.

#### 3.2 Relationship Map (Join Paths)

Primary sales backbone:

customer → invoice → invoiceline → track

Attribution chain:

track → album → artist

Additional relationships:

- track → genre
- track → mediatype
- playlisttrack links playlist ↔ track

## 4 Integrity and Data Quality (Verified)

Profiling checks confirm:

- **Uniqueness:** key IDs are unique (invoice, invoice line, track, customer).
- **Completeness:** invoice date and totals are always present; line unit price and quantity are always present.
- **Referential integrity:** no missing links across audited foreign keys.

This makes revenue attribution and dimensional rollups stable and reproducible.

## 5 Key Profiling Results (Stakeholder-Readable Metrics)

### 5.1 Time and Revenue

- Date range: 2021-01-01 → 2025-12-22
- Invoices: 412
- Revenue: 2328.60
- Average invoice value: 5.65

### 5.2 Geography

- Billing countries: 24
- Top markets by revenue: USA (523.06), Canada (303.96), France (195.10), Brazil (190.10), Germany (156.48)

### 5.3 Customer Patterns

- Invoices per customer: min 6, max 7, avg 6.98
- LTV range: 36.64 to 49.62 (avg 39.47; median 37.62)
- Top-10 customers represent 19.38% of total revenue

### 5.4 Catalog and Pricing

- Track unit price: 0.99 to 1.99 (avg 1.05)
- Invoice line quantity: always 1 (2240/2240 lines)
- Track duration (ms): median 255634; 90th percentile  $\approx$  482992

## 6 Analytical Feasibility

### 6.1 Strongly Supported

- Revenue trends (monthly/yearly)
- Revenue by country and market comparisons
- Top customers / customer contribution
- Artist/genre performance ranking and mix analysis
- Concentration metrics (Pareto-style breakdowns)

### 6.2 Not Supported / Weak

- Churn / retention modeling (no one-time customers)
- Discount / promo impact (no discount data)
- Basket-size scaling by quantity (quantity constant)

## 7 Next Step

Define a small set of **analysis objectives** (business questions and success metrics) aligned with what the data can actually support. Those objectives will drive the design of `marts` tables and the final analysis outputs.